HiPerGator SAM3 + BioEncoder Workflow

This guide walks through setting up and running a SAM3 (Segment Anything 3) segmentation workflow on HiPerGator using a Conda environment. It includes common errors and how to fix them.

▼ HiPerGator SAM3

Phase 1 — Upload Data

From your local machine (e.g., Mac/Linux terminal), upload files using `scp`:

```
scp -r "/local/path/M" USER@hpg.rc.ufl.edu:/blue/.../your_folder/
scp -r "/local/path/F" USER@hpg.rc.ufl.edu:/blue/.../your_folder/
```

To upload CSVs:

```
scp "/local/path/file.csv" USER@hpg.rc.ufl.edu:/blue/.../your_folder/
```

Phase 2 — Start Interactive Session (WITH GPU)

1. Go to Open OnDemand
2. Launch **HiPerGator Desktop**
3. ⚠️ Request **1 GPU** (required for SAM3)
4. Open **Terminal Emulator** inside the session

Phase 3 — Set Up Conda Environment

Load Conda (required on HiPerGator):

```
module load conda
```

Create environment (Python 3.10 recommended):

```
conda create -n sam3_env python=3.10 -y
```

Activate:

```
conda activate sam3_env
```

Phase 4 — Install Dependencies

Install core packages:

```
pip install autodistill autodistill-sam3 roboflow inferenc
e opencv-python numpy scikit-learn sam3
```

Phase 5 — Organize Files (Optional)

If organizing files based on CSV:

```
nano organize_folders.py
```

- Paste script
- Save: `Ctrl + O`, Enter
- Exit: `Ctrl + X`

Run:

```
python organize_folders.py
```

Phase 6 — Set Roboflow API Key

```
export ROBOFLOW_API_KEY="YOUR_KEY"
```

⚠️ Must be done every session unless added to `.bashrc`

To get your API Key

- Go to https://app.roboflow.com/settings/api
- Create or sign in to your account
- Go to Settings → API Keys
- Click on "generate" under "**Private API Key"**
- Copy the key and paste it to your code

Phase 7 — Run Segmentation Script

Create script safely:

```
nano sam3_segmentation.py
```

Paste code → Save → Exit

Run:

```
python sam3_segmentation.py
```

⚠️ Common Errors & Fixes

1. `conda: command not found`

```
module load conda
```

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2. `SyntaxError` with `{\rtf1...`

Cause: File saved as RTF (Mac TextEdit)

Fix:

- Delete file
- Recreate using `nano`

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3. `ModuleNotFoundError: autodistill`

```
pip install autodistill
```

---

4. `ModuleNotFoundError: sklearn`

```
pip install scikit-learn
```

---

5. `ModuleNotFoundError: sam3`

```
pip install sam3
```

---

6. Missing modules (general rule)

If you see:

```
ModuleNotFoundError: <module>
```

Run:

```
pip install <module>
```

---

7. Roboflow API Error (401 Unauthorized)

Error:

```
Unauthorized access to roboflow API
```

Fix:

1. Get new **Private** API key from Roboflow
2. Re-export:

```
export ROBOFLOW_API_KEY="NEW_KEY"
```

---

8. Running outside correct directory

Navigate to working directory:

```
cd /blue/.../your_folder/
```

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9. GPU-related crashes

Cause: Session started without GPU

Fix:

- Restart session
- Request GPU

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10. Broken environment

Reset everything:

```
conda deactivate
conda env remove -n sam3_env
```

Then restart from Phase 3

Tips

- Always use `nano` to create Python scripts on cluster

- Keep all files in `/blue/...` (not home directory)
- Install missing packages incrementally
- First run may take longer (downloads model weights)

Minimal Workflow Summary

```
module load conda
conda create -n sam3_env python=3.10 -y
conda activate sam3_env
pip install autodistill autodistill-sam3 roboflow inferenc
e opencv-python numpy scikit-learn sam3
export ROBOFLOW_API_KEY="YOUR_KEY"
python sam3_segmentation.py
```

▼ BioEncoder Classification

1. Prepare Dataset Structure

BioEncoder expects:

```
dataset/
  class1/
  class2/
  class3/
```

Example:

```
mkdir -p dataset/class1 dataset/class2 dataset/class3
cp segmented_class1/* dataset/class1/
cp segmented_class2/* dataset/class2/
cp segmented_class3/* dataset/class3/
```

2. Install BioEncoder

```
pip install bioencoder
```

---

3. Configure Workspace

```
python -c "import bioencoder; bioencoder.configure(root_dir='bioencoder_wd', run_name='run_v1')"
```

---

4. Split Dataset

```
bioencoder_split_dataset --image-dir "dataset" --max-ratio 6 --random-seed 42
```

---

5. ⚠️ CRITICAL — Fix Number of Classes

Edit BOTH files:

```
nano yml_files/train_stage2.yml
nano yml_files/swa_stage2.yml
```

Set:

- `num_classes: N`
- `classes: N`

Where **N = number of folders/classes**

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6. Train Model

```
bioencoder_train --config-path "yml_files/train_stage1.yml"
```

```
bioencoder_swa --config-path "yml_files/swa_stage1.yml"

bioencoder_interactive_plots --config-path "yml_files/plot_stage1.yml"

bioencoder_train --config-path "yml_files/train_stage2.yml" --overwrite
bioencoder_swa --config-path "yml_files/swa_stage2.yml" --overwrite
```

---

7. Common BioEncoder Errors

CUDA device-side assert triggered

Cause:

- Wrong `num_classes`

Fix:

- Match YAML with dataset folders

FileExistsError

Fix:

```
--overwrite
```

SWA IndexError

Cause:

- Training failed → no weights

Fix:

- Fix training first, then rerun SWA

Checkpoint not found (epochXX)

Fix:

- Use `swa` instead of specific epoch (e.g. “epoch99”)

---

dconf / emacs warnings

- Ignore (harmless on HPC)

Phase 11 — Model Explorer (GradCAM)

1. Fix Config Bug (img_size missing)

```
echo "img_size: 224" > temp.yml
cat yml_files/explore_stage2.yml >> temp.yml
mv temp.yml yml_files/explore_stage2.yml
```

---

2. Fix img_size Format Bug

```
sed -i 's/img_size: \[224, 224\]/img_size: 224/g' yml_files/explore_stage2.yml
```

---

3. Launch Explorer

```
bioencoder_model_explorer --config-path "yml_files/explore_stage2.yml"
```

Open in browser:

```
http://localhost:8501 #example link
```

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4. What You Get

- GradCAM
- Contrastive GradCAM
- Interactive exploration

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5. Common Errors

Image resize crash

Cause:

- `img_size` as list

Fix:

- Use single integer (224)

Nothing loads

- Check Stage 2 completed
- Check weights exist in:

```
bioencoder_wd/weights/
```

Final Notes

- Stage 2 = your real classifier
- GradCAM works only after Stage 2
- Small datasets → unstable metrics (normal)
